

ℵ₀ Weekly Problem

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2025-11-30

Problem

Let a_n be the number of terms in the sequence $2^1, 2^2, \dots, 2^n$ which begin with digit 1. Prove that

$$\log 2 - \frac{1}{n} < \frac{a_n}{n} < \log 2,$$

and, hence,

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \log 2 \approx 0.30103.$$

Solution

Proof. We first prove the following lemma, greatly simplifying the remainder of the proof.

Lemma 1. Define $(s_m) = (2^j)_{j=1}^m$, the sequence of the first m integer powers of 2. Also, let S_c be the set of such sequences all having exactly $c \in \mathbb{N}$ terms whose leading digit is 1. If $(s_m) \in S_c$, then $c \log_2 10 < m < (c+1) \log_2 10$.

Proof. Suppose that $(s_m) \in S_c$ for some $c \in \mathbb{N}$. We will show that

$$c \log_2 10 < m < (c+1) \log_2 10.$$

First, observe that for 2^q to have leading digit 1, there must exist some $t \in \mathbb{N}$ such that

$$10^t \leq 2^q < 2 \cdot 10^t \implies t \log_2 10 \leq q < t \log_2 10 + 1.$$

For fixed t , the interval $[t \log_2 10, t \log_2 10 + 1)$ has length 1, so it contains at most one integer q .

Let $j_1 < j_2 < \dots < j_c \leq m$ be the exponents of the terms of (s_m) that have leading digit 1. For each i there is some t_i with

$$t_i \log_2 10 \leq j_i < t_i \log_2 10 + 1,$$

and the intervals are mutually disjoint, so $t_1 < t_2 < \dots < t_c$. Also, $t_1 \geq 1$ since $2^4 = 16$ is the first positive integer power of 2 with 1 as its most significant digit

in our sequence $(2^j)_{j \geq 1}$. Because $(t_i)_{i=1}^c$ is strictly increasing, we have $t_i \geq i$ for all $1 \leq i \leq c$, and hence

$$j_i \geq t_i \log_2 10 \geq i \log_2 10.$$

In particular, since $\log_2 10$ is irrational, we get

$$m \geq j_c > c \log_2 10,$$

so $m > c \log_2 10$.

To establish the upper bound, we first show that for every $t \in \mathbb{N}$ there is *exactly one* exponent $q_t \in \mathbb{N}$ such that 2^{q_t} has leading digit 1 and lies in the decade $[10^t, 10^{t+1})$. Fix $t \in \mathbb{N}$ and define $q_t := \lceil t \log_2 10 \rceil$. Then

$$t \log_2 10 \leq q_t < t \log_2 10 + 1,$$

so exponentiating base 2 gives

$$10^t \leq 2^{q_t} < 2 \cdot 10^t,$$

which shows that 2^{q_t} begins with digit 1 in base 10. Thus, for each t there exists at least one such exponent q_t , and by the “at most one integer per interval” argument above, this q_t is unique.

It follows that the sequence of all exponents $q \geq 1$ such that 2^q has leading digit 1 is precisely $(q_t)_{t \geq 1}$, ordered increasingly, and the corresponding interval indices are $t_1 = 1, t_2 = 2, \dots$. In particular,

$$t_i = i \quad \text{for all } 1 \leq i \leq c+1.$$

Therefore, $j_c \in [c \log_2 10, c \log_2 10 + 1)$ and $j_{c+1} \in [(c+1) \log_2 10, (c+1) \log_2 10 + 1)$. Since $m < j_{c+1}$ and m, j_{c+1} are integers, we have

$$m \leq j_{c+1} - 1 < (c+1) \log_2 10,$$

which proves $m < (c+1) \log_2 10$.

Combining the two inequalities yields

$$c \log_2 10 < m < (c+1) \log_2 10,$$

as desired. ■

Turning our attention to the original problem, it is sufficient to show that

$$a_n \log_2 10 < n < (a_n + 1) \log_2 10$$

for all $n \in \mathbb{N}$ because

$$\begin{aligned} \log 2 - \frac{1}{n} < \frac{a_n}{n} < \log 2 &\iff n \log 2 - 1 < a_n < n \log 2 \\ &\iff \frac{n}{\log_2 10} - 1 < a_n < \frac{n}{\log_2 10} \\ &\iff a_n \log_2 10 < n < (a_n + 1) \log_2 10. \end{aligned}$$

Let $n \in \mathbb{N}$ be arbitrary. Then, (s_n) has a_n terms whose leading digit is 1 and $(s_n) \in S_{a_n}$. By Lemma 1,

$$a_n \log_2 10 < n < (a_n + 1) \log_2 10,$$

as desired. Since n is arbitrary, we can write

$$\forall n \in \mathbb{N}. \log 2 - \frac{1}{n} < \frac{a_n}{n} < \log 2.$$

By the Squeeze Theorem, we immediately obtain

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \log 2 \approx 0.30103.$$

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